Bandit-Based Material Screening

Project 5, 25 Real-World RL Homework Projects (Unit 5) | Problem formulation

Original problem statement

> *"A multi-armed or contextual bandit that chooses which candidate alloys or compounds to 'test' next from a public dataset (e.g., Matbench)."*

This project uses the multi-armed variant. Scenario: a researcher has a fixed testing budget and must choose which steel to test next, balancing exploration (testing uncertain steels) with exploitation (re-testing steels that look strong).

Formulation

Environment	A pool of 60 candidate steels sampled (fixed seed 42) from the matminer *steel_strength* dataset (312 steels, experimental yield strength in MPa). Each steel is one arm (K = 60) with a hidden true mean strength μ_a. Testing a steel returns a noisy measurement. The primary target property is **yield strength (MPa)**, where higher is better.
State	Stateless multi-armed bandit. The agent's information state is its test history, summarised by counts N(a) and mean estimates Q(a) (plus posterior mean and variance for Thompson Sampling).
Action	a_t in {1, ..., K}: choose which steel to test at experiment t.
Reward	$r_t = \mu_{a_t} + \varepsilon_t$, with $\varepsilon_t \sim \text{Normal}(0, 100^2)$ in MPa. The observed reward is a noisy measurement, never the true property itself.
Horizon	T = 300 experiments (the testing budget), identical for every algorithm.
Objective	Maximise cumulative reward $\Sigma_t r_t$, equivalently minimise cumulative regret $R_T = \Sigma_t (\mu^* - \mu_{a_t})$, and identify the best steel with few tests (μ^* = 2,487 MPa).

Assumptions

- The reported yield strength of each steel is treated as its hidden true mean. The dataset has no replicate measurements, so this is a modelling assumption.
- Measurement noise is simulated: Gaussian, independent across tests, same SD (100 MPa) for all arms. The value is an assumption, not estimated from data.
- **No information leakage:** agents only see noisy rewards through the environment. True means are used only by the evaluator.
- Fair comparison: for each seed all algorithms face the same noise table (common random numbers) and the same pool and budget. 30 independent seeds (0 to 29).
- Arms are independent. Composition features (a contextual bandit) are left as future work.

Algorithms and metrics

Algorithm	Rule
Random	Pick a steel uniformly at random (baseline).
Epsilon-Greedy	With probability 0.1 test a random steel, otherwise the one with highest Q(a). Update: $Q \leftarrow Q + (r - Q) / N$.
UCB1	$\text{argmax}_a [\, Q(a) + c \sqrt{(2 \ln t / N(a))} \,]$, rewards scaled to [0, 1] by 2500 MPa. c = 1 (standard); an extra variant with c = 0.1 is also reported.
Thompson Sampling	Gaussian posterior per steel (prior mean 1000, SD 500 MPa, known noise SD). Sample one value per steel and test the largest.

Metrics: cumulative reward, average reward per test, cumulative regret, simple regret of the recommended steel, best-steel identification rate, tests until a top-10% steel is first tested (top 10% = 6 steels, at least 1,798 MPa), and exploration versus exploitation behaviour.